MILITARY SPECIFICATION

ENGINEERING SERVICES, CALIBRATION EQUIPMENT, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers general requirements for engineering studies; preparation of engineering documentation; fabrication, testing, and evaluation of engineering prototypes; and related product engineering for calibration equipment.

1.2 Classification. The engineering services covered by this specification shall be of the following types:

Type I Engineering Service—Accomplishment of an engineering study and the preparation of all engineering documentation. Consists of compliance with the requirements specified under Phase I of this specification.

Type II Engineering Service—Accomplishment of Type I Engineering Service plus the fabrication and testing of an engineering prototype. Consists of compliance with the requirements specified under Phase I and Phase II of this specification.

Type III Engineering Service—Accomplishment of Type I and Type II Engineering Services plus the production of additional quantity to be specified in the contract.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L–P–517 —Plastic Sheet, Scribe-Coated.

L–P–519 —Plastic Sheet, Tracing, Glazed and Matte Finish.

MILITARY

MIL–D–5480 —Data, Engineering and Technical; Reproduction Requirements for.

MIL–M–9868 —Microfilming of Engineering Documents, 35 mm, Requirements for.

MIL–I–45208 —Inspection System Requirements.

MIL–D–70327 —Drawings, Engineering and Associated Lists.

STANDARDS

FEDERAL

FED-STD No. 5—Standard Guide for Preparation of Item Descriptions by Government Suppliers.

MILITARY

MIL–STD–7 —Types and Definitions of Engineering Drawings.

MIL–STD–16 —Electrical and Electronic Reference Designations.

MIL–STD–31 —Numbering and Cod- of Engineering Drawings, Associated Lists and Documents.

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MIL–STD–143 —Specifications and Standards, Order of Precedence for the Selection of.
MIL–STD–804 —Formats and Coding of Tabulating and Aperture Cards for Engineering Data Micro-Reproduction System.
MIL–STD–1243 —Engineering Drawings.
MIL–STD–1244 —Lists Associated with Engineering Drawings.
MIL–STD–1245 —Engineering Directives and Records.

PUBLICATIONS

DEFENSE STANDARDIZATION MANUAL

M200, Chapter V —Outline of Form and Instructions for the Preparation of Specifications and Associated Documents.
RA NMPC No. 1A—Instructions for Preparation of Manuscripts.
RA NMPC No. 1B—Instructions for Preparation of Illustrations.
RA PUB 1C —Instructions for Typing of Manuscripts.
RA PD No. 29 —Technical Publications for Ordnance Materiel.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Phase I—engineering study. The contractor shall accomplish an engineering analysis of the requirements of the Engineering Description for the item and shall perform such studies and investigations as necessary in the selection or design of equipment to satisfy these requirements. The proposed equipment shall comprise the minimum number necessary for optimum performance.

3.1.1 *Engineering analysis.* The contractor shall prepare an engineering analysis of the dimensions, tolerances, performance, and material requirements of the Engineering Description. The analysis shall include a complete itemized list of all equipments and associated components necessary to satisfy these requirements and should be based on and in the order of the following:

(a) Those items in the supply system (Army), preferably those that have been nomenclatured and standardized,
(b) Items that are in the Air Force or Navy supply system.
(c) Items commercially available.
(d) The development of new equipment based on prepared specifications.

The list shall be accompanied by complete specifications for items that are commercially available. When commercial equipment is not available or does not fully satisfy the requirements of the Engineering Description, the contractor shall make design recommendations as necessary concerning the development of adequate equipment. The contractor shall provide diagrams that clearly delineate the proposed equipments, associated component parts, and their interconnections. Upon Government approval of the design recommendations, the contractor shall determine and establish all necessary performance characteristics for the new developments. Unless otherwise specified, the contractor shall determine the feasibility of any diagrams, sketches, or other design data included with the Engineering Description, ascertaining and providing additional engineerng data where necessary.

3.1.2 *Design studies.* When it is found necessary to design new equipment or components in order to satisfy the requirements of the Engineering Description, the contractor shall provide a design study for each

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item. The study shall contain but not be limited to:

(a) Index of contents.

(b) Introduction, philosophy, theory of operation, and relationship to the pertinent item.

(c) Diagrams showing basic concepts, circuitry, optical systems, etc.

(d) Design layouts drawn to a fixed scale in third angle orthographic projection and in sufficient detail to provide guidance to detail draftsmen. Adequate views shall be shown to clearly depict the size, shape, and operational characteristics of the parts involved and shall include all information necessary for a design analysis.

(e) Theoretical analysis, performance specifications, accuracy requirements, and reliability data upon which design is based.

(f) Operating procedures, giving tentative set-up and operating instructions.

(g) List of operational accessories and associated equipment.

(h) Calibration procedures, where applicable.

3.1.2.1 *Design objectives.* Design objectives to be considered in the study should strive to achieve the following:

(a) Optimum producibility coupled with low unit cost.

(b) Maximum reliability and long service life consistent with reasonable economic factors.

(c) Ease of maintenance, installation, operation, and calibration.

(d) Minimum use of critical or excessively costly materials.

(e) Maximum use of Military Standard or commercial parts to minimize design effort.

(f) Elimination, to the most practical extent, of proprietary items or sole source items.

(g) Optimum man-machine functions.

3.1.2.2 *Human engineering.* In order to realize an optimized man-machine system, the contractor shall apply established principles of human engineering in the selection or design of the equipment or system under consideration during all phases of work required by the Engineering Description. Consideration shall be given, but not limited to, the following factors:

(a) Environmental factors such as temperature, humidity, noise, shock, vibration, dust, illumination, etc.

(b) Human space limitations for operation and maintenance.

(c) Visual and auditory perceptual requirements.

(d) Human safety factors.

(e) Operator duties, skills, educational levels, and training time.

(f) Psychomotor requirements.

3.1.3 *Use of standard and existing parts.* Components selected for incorporation into the equipment design shall, insofar as possible, be standard military components and shall follow the precedence criteria of MIL–STD–143 and MIL–STD–1243 (Ord). Maximum design freedom will be permitted in the selection of components of commercial origin or in the design of special parts, provided that such selection or design is consistent with the requirement that maximum use be made of existing designs and parts currently in the military system. While the introduction of additional parts into the military supply system which offer only marginal improvement over existing or standard parts shall be entirely avoided, recommendations for the introduction of new items which offer marked improvement in performance, reliability, maintenance, or reduced cost, is encouraged and will be favorably considered.

3.1.4 *Equipment justification.* Should the contractor consider that the use of certain commercial, modified commercial, or special parts, in lieu of similar standards or existing parts, will offer marked improvement in performance, reliability, maintenance, or reduced cost, he shall request approval from

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the Government representative prior to incorporation of such items in the final design. Requests for approval should contain the following information:

(a) A description, sketch, or reproducible print.

(b) A comparison of significant characteristics with those of similar existing or standard parts.

(c) Justification for substitution.

(d) Proof (if a commercial or modified commercial part) that item is on the commercial market, available to the public, and is advertised in current catalogs or trade literature at an established price range.

(e) A competent opinion as to Government reprocurement rights if item is patented or otherwise proprietary.

3.1.5 *Component identification.* All electrical and electronic equipment of new design shall incorporate component reference designations in accordance with MIL–STD–16 for marking part locations. Reference designation markings shall be permanent and legible. Wherever practical, marking shall be extended to commercial equipment. Where this will raise the price of commercial equipment above 10% of their quoted stock price, the contracting officer shall have the prerogative of deeming the requirement impractical; unless otherwise stated, it shall be included.

3.1.6 *Safety measures.*

3.1.6.1 *Mechanical.* Proper measures shall be taken in the design of the equipment to prevent accidental contact with moving parts by personnel. Projecting and overhanging edges shall be held to a minimum. Edges and corners of enclosures shall be rounded.

3.1.6.2 *Electrical.* With the equipment assembled and set up for operation, personnel shall be protected from contact with dangerous electrical potentials. Interlocks shall be provided where necessary to prevent access to the interior of the equipment. Maximum protection shall be provided operating and other personnel from dangerous RF and X-ray emissions.

3.1.7 *Proprietary designs and processes.* Proprietary items, processes, circuits, and other inventions shall not be incorporated in the design without advance disclosure of such intent by the contractor, and advance approval by the Government Representative and/or authorization to said rights of this data for use in future Government contracts.

3.1.8 *Convenience.* Each artice of equipment and each major sub-assembly, forming a part thereof, shall provide easy and ready access to its interior parts, terminals, and wiring for adjustments, complete circuit checking, and the removal and replacement of component parts. As a general rule, it will not be acceptable to remove or displace wires, cables, parts, or subassemblies in order to gain access to terminals, parts, mounting screws and the like. When it is not practical to avoid such construction, those parts which must be displaced or removed shall be so designed, mounted, or otherwise arranged to facilitate their displacement or removal when necessary. If, in order to check or remove a given part, it is necessary to displace another part, the latter part shall, whenever practical, be so wired and mounted that it can be sufficiently moved without being disconnected from the circuit.

3.1.9 *Submission of engineering studies.* Engineering studies shall be submitted to the Government representative for approval prior to the preparation and submission of engineering documentation. If engineering studies are rejected on the basis of nonconformace with this specification or the Engineering Description, the contractor shall rework and resubmit new studies on a continuing basis until approval is obtained.

3.1.9.1 *Partial submission.* Engineering studies may be submitted as a complete study or as a portion of the overall project, if breakout of any portion of the project is practical. All breadboards and mockups considered necessary to demonstrate feasibility shall be made available to the Government

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Representative at the time of submission of the engineering study.

3.1.9.2 *Acceptance of engineering studies.* Acceptance by the Government of submitted studies constitutes acceptance of general features such as size, arrangement, apparent feasibility, etc. Approval of engineering studies does not, however, relieve the contractor of the responsibility for compliance with the requirements specified in the Engineering Description.

3.2 Engineering documentation.

3.2.1 *Engineering drawings.* The contractor shall, upon approval of the engineering study, prepare a complete set of engineering drawings for all items developed under the contract and for all commercial items not presently covered by official Government documentation. These drawings shall be in conformance with the requirements of Specification MIL–D–70327 and MIL–STD–1243 (Ord) and all of the specifications, standards, and publications referenced therein as applicable. The contractor shall, to the maximum extent possible, assure that no duplicate drawings will be prepared.

3.2.1.1 *Class of drawings.* Unless otherwise specified in the contract all drawings prepared by the contractor shall be Class 1 (MIL–D–70327) drawings. Class 2 (MIL–D–70327) drawings will be accepted only if such drawings are shown to be in existance prior to issuance of the Invitation for Bids and meet the design disclosure requirements of MIL–D–70327 and the requirements of Grade 4, MIL–STD–1243 (Ord). The acceptance of Class 2 drawings, other than those specifically proposed by the contractor at the time of submission of his bid or proposal, will be contingent upon an equitable adjustment of contract price or fixed fee.

3.2.1.2 *Types of engineering drawings.* Selection of the types of engineering drawings shall be governed by MIL–STD–7, and shall be suitable for:

(a) Unrestricted competitive fixed-price procurement as a complete item.

(b) Unrestricted competitive fixed-price procurement of all components and subassemblies, with the exception of those items covered by "source control" drawings as defined in MIL–STD–7, which have been approved by the Government Representative.

(c) Effective production, process and assembly-line planning.

(d) Determination of the acceptance or rejection of the major item and all components.

(e) Maintenance and troubleshooting purposes as evidenced by adequate assembly and sectional views, verbal descriptions of assembly adjustment procedures, and adequate schematic and diagrammatic drawings for electrical circuitry, piping, etc.

3.2.1.3 *Undimensioned master drawings.* The contractor shall furnish, as part of the set of engineering drawings, all undimensioned master drawings required for the manufacture of etched circuit boards, panels, reticles, and special scales. Master drawings shall be prepared on dimensional stable plastic sheet in accordance with Specification L–P–517 or L–P–519. Drawings shall be of high quality, clearly drawn, typed, or scribed with sharp, clean-cut lines, to insure a high degree of legible reproducibility. Appliqued patterns shall be photographically opaque except for desired open areas. Unless otherwise specified in the contract, the contractor shall supply plastic sheet for master drawings in accordance with standard format requirements.

3.2.1.4 *Drawing formats.* Except as provided in 3.2.1.3, preprinted drawing formats for use in the preparation of Class 1 drawings will be supplied by the Government upon application by the contractor. The application should be accompanied by an estimate of the number of each size of drawing format which will be required.

3.2.1.5 *Drawing numbers.* A block of drawing numbers will be provided for use on the

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Class 1 drawings by the Contracting Officer upon request by the contractor. Appropriate request forms will be supplied by the Contracting Officer. Drawing number cards issued by the Government shall be carefully safeguarded and controlled by the contractor against loss. The contractor shall enter the required nomenclature and application data on each card and shall return all used and unused drawing number cards to the Government upon completion of the contract or at such earlier time as may be specified.

3.2.1.6 *Code identification.* The proper code identification number shall be placed on all Class 1 drawings in conformance with the requirements of MIL–STD–31.

3.2.1.7 *Preparation of drawings.* Class 1 drawings prepared by the contractor shall be Grade 2, per MIL–STD–1243 (Ord) and shall be sufficiently legible to produce a fourth generation reproduction in accordance with Specification MIL–D–5480.

3.2.2 *Engineering lists.* The contractor shall prepare the following engineering lists in accordance with the requirements of MIL–STD–1244 (Ord). They shall be prepared separate from the drawings using blank forms to be supplied by the Contracting Officer:

(a) Parts List (PL)—one required for each major item and for each intermediate assembly consisting of two or more separable parts.

(b) Data List (DL)—one cumulative list required for each major item, for each component or intermediate assembly.

(c) Equipment List (EL)—one required for each combination of a major item and necessary ancillary equipment, such as cables, adapters, carrying case, etc.

(d) Index List (IL)—one required for each system comprised of two or more major items; and for each combination of a major item with equipment.

3.2.3 *Military specifications.* The contractor shall prepare a military specification in accordance with chapter V, Defense Standardization Manual M200, and this specification, for the major item described in the Engineering Description and for such components and intermediate assemblies as determined by the contractor or as specified by the Government representative. A list of proposed component specifications shall be submitted to the Government representative for concurrence prior to the preparation of the specifications. Magnitudes, rates, limits, etc. set forth in the specifications shall be capable of being measured with either commercially available equipment or equipment available to the Government. The contractor shall be responsible for editorial and technical accuracy of the specifications.

3.2.3.1 *Preparation of section 3 of military specifications.* The following instructions for preparation of section 3 of Military specifications shall supplement the requirements of Chapter V, Standardization Manual M200:

3.2.3.1.1 *Preproduction model.* A requirement for a preproduction model will normally be included in section 3 of all specifications except when a qualification requirement is specified or when the design of item is covered by referenced drawings. A requirement for a preproduction model will not normally be specified in section 3 of specifications which reference such drawings, exxcept in cases where special tests are desired over and above those which are practical for normal inspection. These will usually require special test facilities or Government proving grounds. A preproduction model should not be specified for the purpose of proofing the adequacy of Government technical data, nor should it be specified in order to determine the contractor's ability to meet contract requirements. Both of these objectives are more appropriately achieved through pre-contract evaluation, or by special provisions of a supply contract.

3.2.3.1.2 *Qualification.* Qualification samples will not be specified as a requirement of section 3 except as specifically directed by the Government representative. Establishment of a Qualified Products List for a

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specific product restricts competition to those suppliers whose product has been tested and approved by the Government prior to procurement, and must be thoroughly justified on the basis of the time and expense involved in conducting special environmental, storage and reliability tests.

3.2.3.1.3 *Specifying requirements.* Section 3 of a submitted specification shall be so written that compliance with all requirements will insure the suitability of the materiel for the intended purpose, and noncompliance with any requirement will indicate the unsuitability of the materiel for the intended purpose. Each requirement shall be necessary, practically attainable, concisely stated and unambiguous. Each requirement shall be self-sufficient and shall be stated in terms independent of test methods and procedures. Section 3 of a specification shall be completely independent of Section 4 and shall serve as the basis for preparation of section 4; not vice versa. Section 3 requirements shall be so specified as to be inherently incapable of being affected by any conceivable change to section 4.

3.2.3.2 *Preliminary drafts.* A preliminary draft of each required specification shall be submitted to the Government representative for review and approval. The draft shall be typed double-space on white bond paper. Unless otherwise specified in the contract, drafts shall be submitted no later than 60 days after approval of design studies.

3.2.3.3 *Final copy.* The final copy of each required specification will be prepared by the contractor based on comment by the Government representative. Copy shall be typewritten, single- space on paper lithographic mats equivalent to Multilith Short-run No. 1 Duplimat Master, Catalog No. 5–3200. Margins shall be suitably positioned for transfer to 8 X 10½" bond paper.

3.2.3.3.1 *Workmanship.* Typing shall be sharp and clean and of uniformly high density. Typewriter ribbons and error correction techniques shall be compatible with the intended lithographic reproduction process. Copy shall be free of overstrikes and typographical errors.

3.2.3.3.2 *Illustrations.* Illustrations used in specifications shall be of the contrast and clarity required for satisfactory reproduction by the lithographic process.

3.2.4 *Release documentation.* The contractor shall prepare an "Initial Release" Engineering Order (EO) for each major item, an Engineering Release Notice for each separate engineering drawing, and an Engineering Order Inclosure List in conformance with the requirements of MIL–STD–1245 (Ord). Engineering Orders and Engineering Notices shall be typed on standard formats using carbon backing, or other suitable method, to obtain maximum density required for reproduction by normal contact printing methods. Standard forms and Engineering Order serial numbers will be issued by the contracting officer upon request.

3.2.5 *Documentation approval.* At such time as the contractor shall consider all or part of the drawings and other documentation to be ready for approval, the originals, together with two reproducible prints of each document shall be reliably delivered to the Government for approval signature and official release. Unless otherwise specified in the contract, the signed original drawings will be returned to the contractor for maintenance for the duration of the contract (see 3.5.1).

3.2.6 *Final report.* A final report shall be submitted to the Contracting Officer within thirty (30) days after completion of Phase I. The report shall contain a summary of the work performed, together with a detailed analysis indicating the basis for acceptance of each design approach developed and evaluated. All substantiating data in the form of notes, calculations, and sketches necessary for its utilization shall be included (see 3.5.8). The report shall also verify that all drawings have been approved. Although no specific format is required, a great deal of care shall be exercised in preparation of the final report. It will become a permanent record and the conclusions, recommendations

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and data contained therein may have future use as a reference or in gathering material in new investigations.

3.3 Phase II—prototype fabrication and evaluation. The following requirements are applicable where the contract or Engineering Description specifically requires the manufacture of an engineering prototype.

3.3.1 *Authorization.* After approval of Phase I, the contractor shall proceed with the fabrication of an engineering prototype from the approved engineering documentation.

3.3.2 *Use of approved engineering documentation.* In order to test the adequacy of drawings and specifications for manufacturing purposes and to assure fabrication in conformance with the intent of the design, the contractor shall manufacture or purchase components for the engineering prototype on the basis of Government approved documentation. Manufacture or purchase of components for which documentation has not yet been approved will be permitted only by written authorization of the Contracting Officer.

3.3.2.1 *Unauthorized documentation.* Technical data, industry or company specifications or standards, not intended to be delivered or disclosed to the Government, shall not be used to supplement the approved drawings or specifications during manufacture, purchase, or inspection. This restriction shall not apply to the use of administrative procedural specifications or standards normally used by the contractor in his manufacturing or purchasing operations.

3.3.3 *Conformance with requirements.* The assembled prototype model shall conform with dimensions, tolerances, materials, finish, and functional requirements specified in approved documentation and the Engineering Description.

3.3.4 *Adjustment.* The contractor shall adjust and align the prototype equipment for optimum performance and then shall measure its performance characteristics.

3.3.5 *Test evaluation.* The contractor shall perform all necessary engineering tests and evaluations of the prototype to determine its conformance to the requirements specified in the engineering description, military documents referenced herein, and specifications prepared under the contract.

3.3.5.1 *Government furnished property.* When equipment or material is furnished by the Government to the contractor for his use in the performance of the contract, the contractor shall be responsible for the following, as a minimum:

(a) Examination upon receipt to detect possible damage in transit.
(b) Inspection for completeness and proper type.
(c) Periodic inspection and precautions to guard against improper handling and deterioration during storage.
(d) Functional testing, either prior to or after installation, or both, to determine satisfactory operation.
(e) Prompt notification to the contracting officer of any delay in receiving Government furnished property or material.

3.3.5.1.1 *Damaged property.* The contractor shall report any Government furnished equipment or material found damaged, malfunctioning, or otherwise unsuitable for use. In the event of damage or malfunction during or after installation, the contractor shall determine and record the probable cause and the reason for withdrawing from use.

3.3.6 *Test evaluation report.* The results of the test and evaluation shall be reflected in a final report which contains as a minimum the following:

(a) An engineering resume of the physical and operational characteristics pertinent to the item.
(b) All sketches, graphs, specifications, drawings, and tabulated performance data necessary for Government evaluation.
(c) A list of all test equipment and detailed procedures used in the

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test and evaluation of each item. Test equipment shall bear a National Bureau of Standards Certification of Calibration. The contractor may submit in lieu of NBS Certification his detailed inspection plan in accordance with MIL–I–45208. This plan shall clearly show that his inspection system has been properly referenced to national reference standards.

(d) A list of accessories required, with specifications for same.

3.3.6.1 *Format.* Although no specific format is required, a great deal of care shall be exercised in the preparation of the final report. It will become a permanent record and the conclusions, recommendations and data contained therein may have future use as a reference or in gathering material for further improvements.

3.4 **Phase III—production of additional quantity.** After approval of Phase II, the contractor shall proceed to the fabrication of the additional production quantity as specified in the contract.

3.4.1 *Product engineering.* The contractor shall perform all product engineering necessary to incorporate any corrective measures or improvements found during prototype evaluation (see 3.3.5). The contractor shall also incorporate all advancements in the state-of-the-art of components and equipments, within the scope of the engineering description, into the production units.

3.5 **General requirements.** Unless otherwise specified in the contract or Engineering Description, the following requirements are applicable regardless of the type of engineering service contracted for.

3.5.1 *Documentation maintenance.* The contractor shall maintain all drawings, specifications, and engineering lists in an up-to-date status for the duration of the contract or until physical custody of the original documentation is assumed by the Government.

3.5.1.1 *Minor revisions.* During the design stabilization period the contractor shall revise technical data to include such changes as may be found necessary during manufacture, inspection, and special testing of the prototype. He may, without prior approval, revise approved Government drawings in his custody, for minor changes in design, dimensioning, tolerances, materials, correction of drafting errors, etc. In such cases the contractor shall prepare and submit to the Government representative a complete Engineering Order (EO) package consisting of one reproducible print of the revised drawing(s), a fully descriptive Engineering Order, an Engineering Order Inclosure List, an Engineering Notice, and an Engineering Change Request, and such parts lists as are applicable in accordance with the requirements of MIL–STD–1245 (Ord) "Engineering Directives and Records." When the contractor has reasonable doubt as to approval of a proposed change, or when it appears to be of a controversial nature, the change shall be discussed with the Government representative prior to the revision of the drawings. Any change which will affect interchangeability, production, or service life will be construed to be a controversial change; other type changes may also fall within this category. In the event of nonconcurrence by the Government representative with the revision, the drawings shall be returned to their previous state or otherwise revised in accordance with instructions of the Government representative. The approved Engineering Order or copy thereof will serve as authorization to accept material conforming with the revised drawing or specification.

3.5.1.2 *Major revisions.* The contractor shall make no changes to any approved drawing or specification which may adversely affect function, service life, maintainability, or human factors, or which will increase the cost of the prototypes by more than 1%. Proposed changes in these categories will be submitted for the consideration of the Government representative by means of an

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Engineering Change Request in accordance with MIL–STD–1245 (Ord) "Engineering Directives and Records." Technical data shall be revised only upon receipt of a copy of the approved request.

3.5.1.3 *Engineering change requests (ECR).* ECR's submitted by the contractor shall be complete with part or drawing change requests, estimated cost data sheets (if required), and sufficient sketches and data as may be necessary to permit evaluation by the Government representative. These request packages shall be transmitted by expeditious means to the Government representative by the contractor. Drawing revisions desired by the Government will be transmitted to the contractor in the form of approved ECR's. Equitable adjustment will be made for change directives originating from the Government which change the scope of the contract requirements. Upon receipt of the approved ECR's, the contractor shall proceed with revision of the affected drawings and preparation of an EO package.

3.5.2 *Record system.* The contractor shall maintain an adequate file of all original EO's, ECR's, lists, and supporting records pertaining to all drawings for which physical custody is assigned. This file shall be returned to the Government at completion of the contract.

3.5.3 *Nomenclature.* The contractor shall furnish a description of each incomplete nomenclatured item in accordance with Federal Standard 5. These descriptions shall be prepared as soon as an adequate definition is available and forwarded to the Government. Upon assignment of approved Military designators, the Government will furnish the contractor with same for inclusion in the engineering drawings.

3.5.4 *Operating instruction manual.* The contractor shall prepare an operating instruction manual to cover all the items which comprise the calibration equipment. This manual shall contain, but not necessarily be limited to, the following:

3.5.4.1 *Operation.* Operating instructions shall include preparation for operation, operation, operation precautions, and removal from operation.

3.5.4.2 *Calibration.* This manual shall contain a section describing the calibration procedures, self-checking or otherwise, necessary to insure satisfactory operation of the equipment. The procedures shall include any additional equipment that may be required and recommended periods for routine calibration.

3.5.4.3 *Maintenance.* Maintenance shall include maintenance procedures, both routine and special, which may have to be performed. These instructions shall include the following, as required:

3.5.4.3.1 *Care.* Care, lubrication, cleaning, sealing and the like. Charts indicating location and frequency are recommended.

3.5.4.3.2 *Assembly.* Assembly and disassembly instructions as are necessary for the maintenance described.

3.5.4.3.3 *Replacement.* Description of replacement and repair when such operations can be performed.

3.5.4.4 *Illustrations.* Illustrations required for operation, maintenance and/or replacement of parts shall be in the form of pictorial views, schematics, and diagrams. Complete circuit schematics shall be presented indicating components, identifying symbols, and component description.

3.5.4.5 *Binder.* This instruction manual shall be incorporated into a suitable binder and adequately titled.

3.5.4.6 *Rough drafts.* Rough drafts shall be submitted to the Government representative for tentative approval of contents, format and accuracy, spaced to facilitate the indicating of changes and/or corrections. Where drawings, sketches, diagrams, or photographs are required, the contractor may substitute either unretouched photographs or sketches when submitting the rough draft copies for approval.

3.5.4.7 *Final drafts.* Final drafts of the operating instruction manual shall be pre-

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pared by the contractor on the basis of the approved rough drafts and comments received from the Government representative.

3.5.4.8 *Negatives.* Should the contractor incorporate photographs in the operating instruction manual, the negatives used to produce the photographs shall be furnished with the final draft. Drawings, sketches, or diagrams incorporated in the operating instruction manual shall be of reproducible quality and shall accompany the final draft.

3.5.4.9 *Guidance for publications.* The contractor shall use the following publications as a guide in the preparation of the Operating Instruction Manual:

3.5.4.9.1 *Format.* RA NMPC Instructions No. 1A shall be used for guidance of technical writers and editors. It prescribes the writing standards, formats, and planning of illustrations, and describes procedures for submitting material for review.

3.5.4.9.2 *Illustrations.* RA NMPC Instructions No. 1B covers the preparation of illustrations by the photographer and illustrator.

3.5.4.9.3 *Typing.* RA PUB Instructions No. 1C describes typing requirements and manuscript arrangement and contains an approved list of abbreviations.

3.5.4.9.4 *Operating manual content.* RA PD–29 shall be used for general requirements to be met in the preparation of an Operating Instruction Manual.

3.5.5 *Microfilm aperture cards.* If specifically required in the contract, the contractor shall supply six, punchcard mounted, 35 mm camera, original microfilm reproductions of each drawing, and engineering list, in accordance with the requirements of Specification MIL–M–9868 and MIL–STD–804. Such microfilm cards shall be furnished by the contractor within a maximum of 21 calendar days after receipt of signed original engineering documents.

3.5.6 *Monthly progress reports.* A narrative progress report shall be submitted monthly during the life of the contract. Five copies, including one reproducible copy, shall be forwarded to the Contracting Officer. While no standard format is required, the report shall include as a minimum the following information:

(a) A resume of the activity of each phase of work during the reporting period.
(b) Percentage completion of each phase of work. (The percentage completion of required documentation shall also be included).
(c) Technical difficulties encountered or anticipated and their solution.
(d) Progress against detailed plan of accomplishment; forecast of accomplishment for succeeding month.
(a) Areas of present or forseeable schedule shippage and planned corrective action.
(f) Expenditures incurred during last reporting period and cumulative expenditures to date.
(g) Estimated funds to be expanded during next reporting period.

The first report shall include a chart showing the detailed plan of accomplishment of each important activity contributing to the successful and timely completion of the contract. In each subsequent report, the chart shall show progress made against each planned activity. Any necessary revision of the plan shall be explained in the narrative report.

3.5.7 *Progress control.* Work done by the contractor shall be suspended immediately on any project activity, or portion thereof, when so requested by the Government. In the absence of such notification, the contractor shall consider the objective, approach and expenditure of funds, as reported in the monthly progress report, to be satisfactory and shall continue toward completion of the specified work.

3.5.8 *Preservation of design data.* It is the responsibility of the contractor to preserve the design notes, calculations, sketches, and other data that precede approval of a design. This preservation shall include all data; that which applies to the proposed design as well as that which applies to the

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design as finally approved. It is preferred that all necessary data be preserved on forms similar to those illustrated in figures 1 and 2. Sketches and other illustrations that cannot be presented on the forms may be presented separately but must be adequately identified with and cross-referenced to the pertinent forms. The type form illustrated in figure 1 shall be used when only one sheet is required to present data on a particular subject; or it shall be used as the first sheet of a multiple sheet presentation with the form in figure 2 being used for subsequent sheets.

3.5.9 *Liaison with the Government.* The contractor is responsible for maintaining adequate liaison with the Government representative to insure prompt and efficient transmission of information and material submitted for approval, including the hand-carrying of such information and material as required. The extent of travel in connection with the overall engineering program will be determined by agreement between the contractor and the Government.

3.5.9.1 *Consultation and assistance by the Government.* Representatives of the Government will be available as required for consultation and assistance pertaining to necessary informational needs of the contractor in the execution of the contract.

3.5.10 *Technical assistance by the Government.* The contractor shall, upon written request from the contracting officer, visit such facilities, agencies and installations as may be designated, for the purpose of participating and assisting in tests and demonstrations, or otherwise acting as agent for, or assistant to the Government, on projects within the scope of the assigned engineering program.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *General provisions.*

4.1.1.1 *Engineering documentation.* The contractor shall provide and maintain an adequate system for the detailed checking and technical review of all analyses, design studies, drawings, lists, and specifications to be supplied under the terms of the contract. A written description of the checking and review system to be used shall be submitted to the contracting officer for approval within 30 days after award of the contract. Changes in the system shall thereafter be made only with the approval of the contracting officer. The minimum requirements to be included in the contractors written checking system shall be as follows:

(a) Organizational assignment of responsibility for detail checking, review, and final contractor approval of engineering documentation.

(b) Resumes of qualification of design checkers, detail drawing checkers, specification reviewers, and personnel having final approval authority.

(c) A description of the procedural flow of all documentation including the extent and depth of review at each echelon of concurrence, review; and approval.

(d) A copy or description of check-lists and guides to be used in the detail checking, review and approval of engineering documentation.

(e) Identification of all characteristics which could result in critical defects, and a resume of the qualifications of the engineer(s) who will attest to the accuracy of critical data.

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DESIGN DATA SUMMARY SHEET	SHEET OF SHEETS
PREPARING ACTIVITY:	CODE, PROJECT, CONTRACT OR FILE NO:
SUBJECT:	ITEM:
	PERTAINS TO:
REFERENCES:	
SUMMARY OF DATA (RECOMMENDATIONS, ACTIONS, CALCULATIONS, RESULTS, ETC.):	
WORK DONE BY _____ DATE	REVIEWED BY _____ DATE

FIGURE 1. *Design data summary sheet.*

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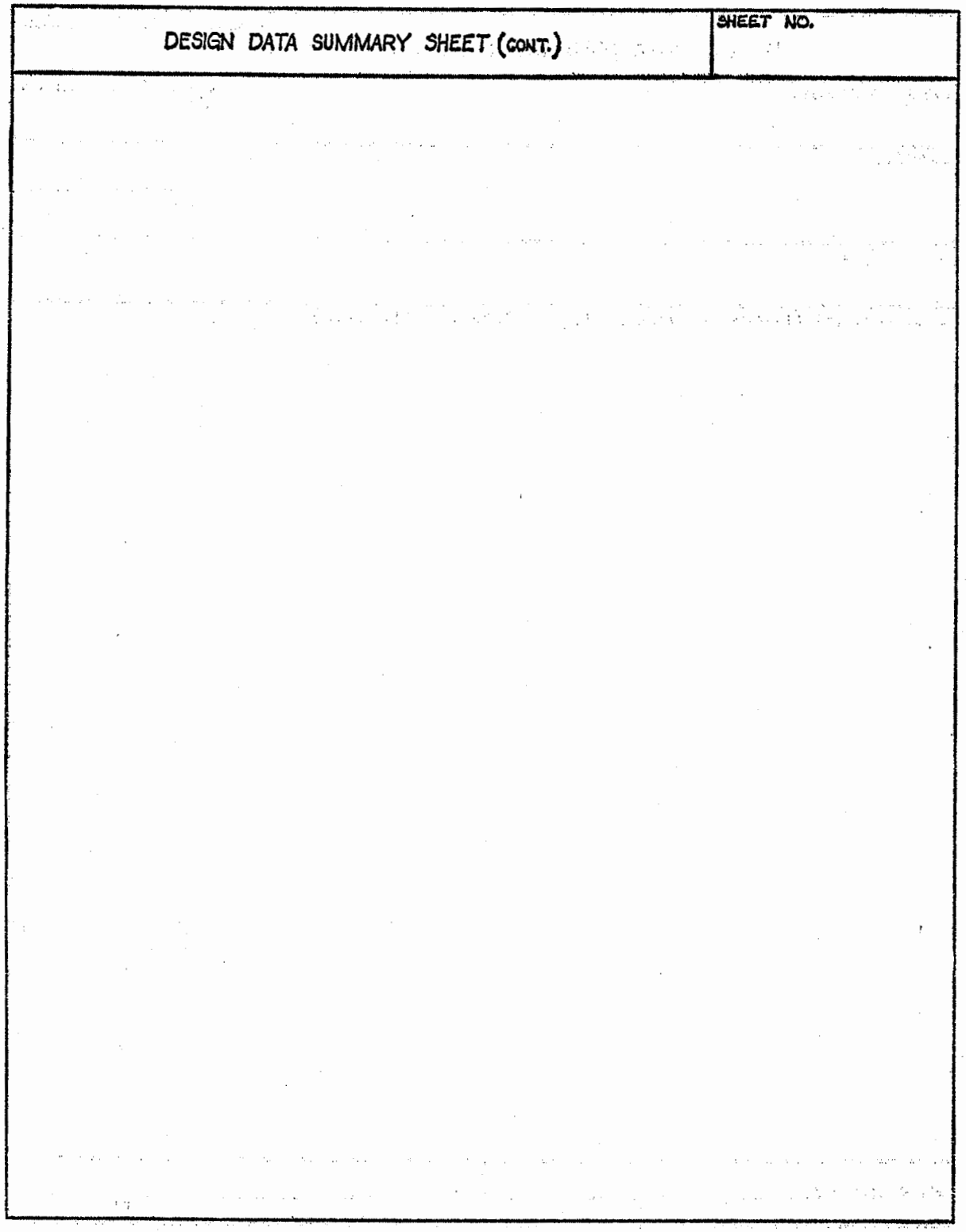


FIGURE 2. *Design data summary continuation sheet.*

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(f) Procedures to be used in sampling inspection of engineering documentation prior to submission to the Government, and a resume of qualifications of personnel who will conduct the inspection.

4.1.1.2 *Engineering prototype.* The contractor shall provide and maintain an adequate inspection system, acceptable to the contracting officer, covering inspection of the components and final testing of the prototype. The contractor shall perform all inspections required by this inspection system. A written description of the inspections system shall be submitted to the contracting officer prior to initiation of prototype manufacture. The written description will be acceptable when it provides as a minimum, assurance that prototypes are inspected and tested in strict conformance with approved documentation and authorized changes thereto. Changes in the system will thereafter be made only with the approval of the contracting officer. The minimum requirements to be included in the contractors inspection system shall be as follows:

(a) An outline summary of the inspection organization, qualifications of inspectors, and management officials to whom they report.
(b) Method of recording inspection results.
(c) Procedures for segregation and disposition of nonconforming material.
(d) Control of engineering changes and requests for deviation approval.

4.2 Inspection provisions for engineering documentation.

4.2.1 *Checking.* The contractor shall conduct a 100% check of all drawings, lists, and specifications prior to submission to the Government for acceptance. Checking shall be conducted by authorized personnel (para. 4.1.1.1) other than the person or persons who prepared the document. The checker's initials on the original drawing, or signature on the cover sheet of a specification, shall constitute a certification of compliance with all applicable requirements of the contractors written checking system, and with all the requirements of this specification and the contract.

4.2.1.1 *Certification of critical data.* The accuracy of technical data in which critical defects (para. 4.2.2.4.1) may occur shall be certified by a qualified engineer.

4.2.2 *Examination.* Technical documentation will be accepted by the Government on the basis of examination of a sampling of each lot of drawings submitted by the contractor. The criteria specified in Table I shall constitute the minimum independent rechecking by the contractor subsequent to the 100% check required by 4.2.1, will constitute the maximum severity of examination by the Government for acceptance or rejection by lot. The Government reserves the right to conduct 100% inspection of a lot or of any portion of a lot, and to reject all individual nonconforming documents. Technical data in which critical defects may occur will be subject to 100% examination by the Government.

TABLE I. *Lot acceptance criteria*

Lot size	Sample	Major defects permitted (AQL 2.5%)	Minor defects permitted (AQL 10%)
1– 5	100% Inspection		
6– 25	5	0	1
25– 40	7	0	1
41– 65	10	0	2
66– 100	15	0	3
101– 200	25	1	4
201– 300	35	2	7
301– 500	50	3	9
501– 800	75	4	13
801–1300	110	6	18

4.2.2.1 *Lot formation.* For purposes of acceptance or rejection by lot, a lot shall consist of any number of related documents prepared and checked under common supervision in a single organization governed by a single set of procedures, policies, and standards. Engineering drawings, engineer-

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ing lists, and specifications shall be submitted as separate lots for examination purposes. The Government reserves the right to combine submitted lots to form a single lot, or to divide a single lot into smaller lots, as may be required to facilitate sampling inspection.

4.2.2.2 *Sampling of engineering drawings, and engineering lists.* Each lot will be arranged in numerical sequence. The frequency of sampling will be determined by the ratio of the size of the lot to the size of the sample as specified in table I. For example, if the lot size is 150, the sample size will be 25, and the ratio of lot size to sample size will be 6; therefore every sixth document will be selected for inspection. The first sample will be selected at random from among the first six documents in numerical sequence, followed by every sixth document in numerical sequence thereafter.

4.2.2.3 *Military specifications.* Military specifications will be accepted on the basis of 100% examination by the Government for conformance with all requirements of this specification and supporting documents.

4.2.2.4 *Classification of defects.*

4.2.2.4.1 *Critical defects.* A critical defect in technical data shall be defined as one, which if undetected prior to use of the technical data, might result in extensive damage to equipment and property, or injury to personnel during manufacture, testing, transportation, or use of the item.

4.2.2.4.2 *Major defects.* A major defect in technical data shall be defined as one which would contribute to one or more of the following deficiencies:

(a) Inability to successfully procure or manufacture the item.
Examples: insufficient description; missing dimensions, tolerances or characteristics; nonexistent part number; unavailable materials or processes; ambiguous or impossible manufacturing conditions or requirements.

(b) Item manufactured or purchased in conformance with data will not assemble or function as intended. *Examples:* incorrect mating dimensions; excessive tolerances; erroneous electrical characteristics; incorrect wire lists; improper selection of materials and finishes, etc.

(c) Data will not support rejection of material, components or major items not suitable for their intended purpose. *Examples:* lack of specificity in detail requirements; lack of adequate inspection or test provisions; ambiguity or conflict between performance requirements and inspection provisions.

(d) Essential processing and asssembly sequence not specified, Examples: insufficient use of subassembly drawings, of section views, and assembly and adjustment instructions; omission of important processing data.

(e) Technical data unduly limits competitive procurement. *Examples:* unauthorized use of proprietary components or processes; unnecessary use of "source control" drawings; specification control drawings unnecessarily "tailored" to a specific source; selection of materials and processes not widely available.

(f) Technical data does not support maximum standardization or interchangeability of components. *Examples:* use of non-standard parts where suitable standards are available; unnecessary use of selective assembly and "fit at assembly" techniques.

(g) Errors in Engineering Parts Lists which can result in procurement or manufacture of incorrect quantities of components. *Examples:* Incorrect part numbers and quan-

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tity requirements; inconsistencies between parts list and assembly drawings or schematics.

(h) Errors in Data List which can result in difficulty and error in compilation of a complete engineering documentation package. Examples: incorrect or omitted drawing numbers; incorrect or omitted specifications; failure to cross-reference supporting lists.

(i) Inability to obtain a completely legible 4th generation reproducible contact print or a completely legible enlarged reproduction from second generation microfilm. *Examples:* insufficient density and contrast in original drafting or typing; insufficient spacing between lines, crowded and insufficiently large lettering; damaged or worn original documents.

4.2.2.4.3 *Minor defects.* A minor defect in technical data shall be defined as any departure from the requirements of this specification and other supporting documents forming part of the contract which are not classifiable as critical or major defects as defined in 4.2.2.4.1 and 4.2.2.4.2. Included in this classification are:

(a) Variations from standard drafting practice which do not materially affect the usefulness of technical data.

(b) Variations from standard format requirements.

(c) Errors in spelling; nonstandard abbreviations; punctuation; and grammatical construction which do not materially affect the clarity of the document.

(d) Departures from accepted standards of neatness and good workmanship which do not affect legibility and clarity of the document.

(e) Use of unnecessarily restrictive tolerances, performance requirements, and materials, provided that manufacture is practical and competitive procurement is not unduly restrained thereby.

4.2.2.5 *Disposition of rejected lots.* Rejected lots shall be screened by the contractor for all defective characteristics. After corrections have been made, the lot shall be resubmitted as a segregated lot, together with a report of corrective action taken. Failure of a lot to qualify for acceptance on a second examination will constitute prima facie evidence of the contractor's inability to perform under terms of the contract.

4.2.2.6 *Disposition of individual nonconforming documents.* Rejected documents from accepted lots will be completely rechecked by the contractor and resubmitted as a single segregated document together with a report of corrective action taken.

4.3 Inspection Provisions for Engineering Prototypes

4.3.1 *Component inspection.* The contractor shall perform 100% inspection of all components on the basis of Government approved drawings and specifications. Any deviations between completed parts and specified requirements shall be recorded and shall be approved by the Government representative prior to submittal of the prototype for acceptance.

4.3.2 *Final inspection.* The contractor shall perform all inspection and tests required by the Government approved specification. Inspection reports of the examinations and tests of all materiel shall be kept complete and available to the Government. Failure of the prototype to meet the functional and performance requirements of the specification, using the prescribed test methods and procedures, shall be cause for rejection of the prototype, pending correction of either the prototype or the specification, or both, as will be determined by the Government representative.

4.3.3 *Contractor certification.* Prior to delivery of engineering prototypes, the con-

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tractor shall certify that all components were manufactured, procured or assembled on the basis of the requirements of Government approved drawings and specifications, and that no other technical data, Industry or company specifications which do not form a part of the final engineering documentation package were used to supplement the drawings and specifications during manufacture, purchase, or inspection. Any exceptions shall be cited in the certification. The contractor shall further certify that all components were given 100% inspection against the requirements of the Government approved drawings and specifications, and that any deviations found to exist were reported to and approved by the Government representative.

4.3.4 *Government inspection.* The Government reserves the right to conduct inspection surveillance or independent inspection of any or all components and assemblies during the course of prototype manufacture and final inspection.

5. PREPARATION FOR DELIVERY

5.1 Engineering documentation. Class I original drawings and Class 2, Grade 4 reproductions shall be carefully handled throughout the drafting, checking, and approval cycle, prior to delivery to the Government. Paper lithographic mats shall be inserted in individual transparent plastic envelopes, in which they shall remain throughout proofreading and in which they will be transmitted to the Government. Preparation for delivery shall be as prescribed in Specification MIL–D–70327.

5.2 Prototype. Packing, and marking shall be in accordance with the contractors best commercial practices to insure safe delivery at destination.

6. NOTES

6.1 Intended use. This specification is intended for use in conjunction with an Engineering Description which will delineate the specific objectives, and performance requirements to be achieved on contracts for the design of calibration equipment. Where substantially less than half of this specification will apply to a contract, the applicable requirements of this specification should be extracted and made an integral part of the engineering description.

6.2 Ordering Data. Use of this specification requires careful delineation in the Engineering Description or the contract, of the type of engineering services desired and of the applicability of each major requirement of this specification, any exceptions or additions thereto, and schedules for completion of each specific requirement. Normally, the Engineering Description will include only technical requirements; delivery data will normally be included in the contract schedule. Where time-phasing of various requirements is critical to timely technical accomplishment, however, schedules may be included in the Engineering Description in terms of time from date of award of the contract.

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Army—MU

Preparing activity:
Army—MU

Project No. MI03–A002

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